Wigner-Crystal to Gas Phase Transition

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Introduction

This study examines a system of N identical ions of charge Q interacting via the Coulomb potential,

$$V(r_{ij}) = \frac{Q^2}{4\pi\epsilon_0 |r_i - r_j|}, \quad (1)$$

confined to a cubic box of volume L^3 with periodic boundary conditions (PBC).

The equilibrium behavior is governed by the competition between electrostatic repulsion and thermal fluctuations, characterized by the coupling parameter

$$\Gamma = \frac{Q^2}{4\pi\epsilon_0 a k_B T}, \quad a = \left(\frac{3}{4\pi n}\right)^{1/3}, \quad (2)$$

where a is the ion-sphere radius and $n = N/L^3$ is the density. At $\uparrow\Gamma$ ($\downarrow T$), ions form a **Wigner crystal** with long-range order, while at $\downarrow\Gamma$ ($\uparrow T$) they behave as a **classical gas**. In the thermodynamic limit this model corresponds to the One-Component Plasma (OCP), for which the long-range Coulomb interaction diverges and typically requires a neutralizing background or Ewald summation [1, 2].

Model and Methodology

We employ the Metropolis Monte Carlo method to sample equilibrium configurations according to the Boltzmann distribution $P \propto \exp(-V/k_B T)$.

- **System:** $N = 216$ point charges in a 3D cubic box with PBC (using the Minimum Image Convention) and no neutralizing background. Reliable for $\Gamma \lesssim 10$ [3], this setup still captures the qualitative features of the phase transition.
- **Simulated Annealing:** Geometric cooling, $T_{i+1} = \alpha T_i$, is used to reach low-energy crystalline states.
- **Melting and Equilibration:** Configurations are reheated to sample multiple temperatures for structural analysis.
- **Stabilization:** Burn-in and equilibration steps are included to ensure thermally stable configurations.

Thermodynamic Properties

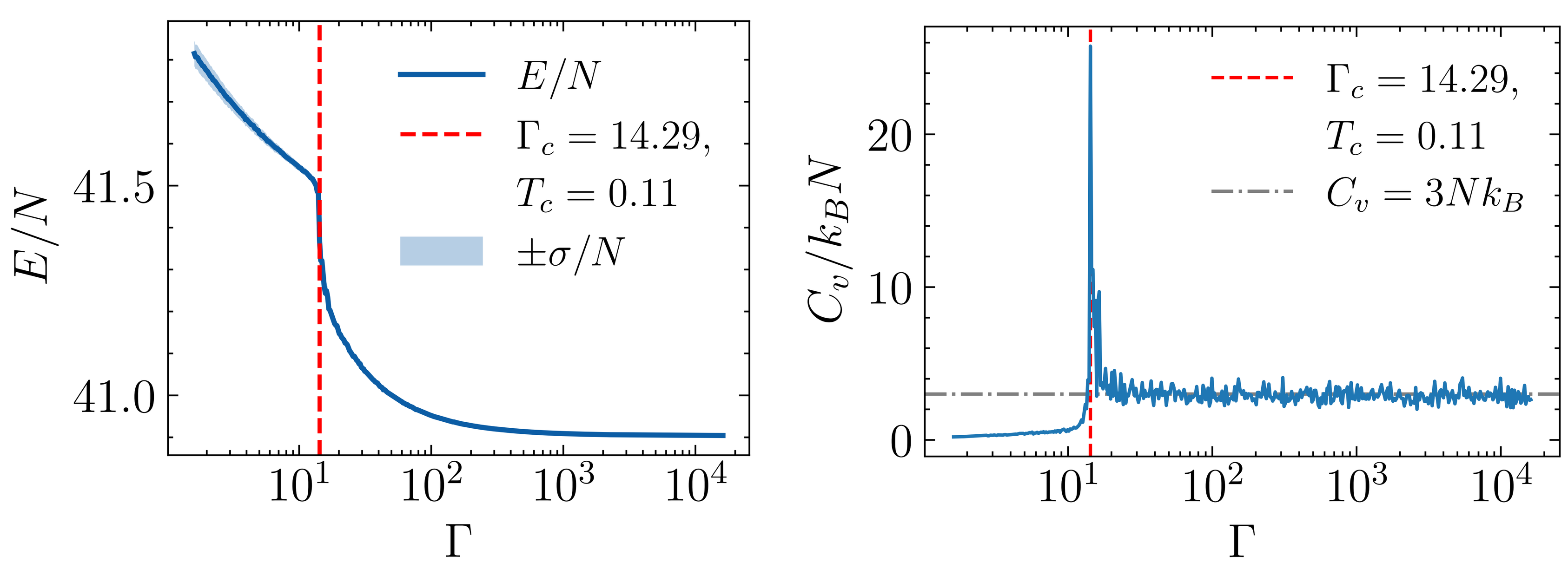
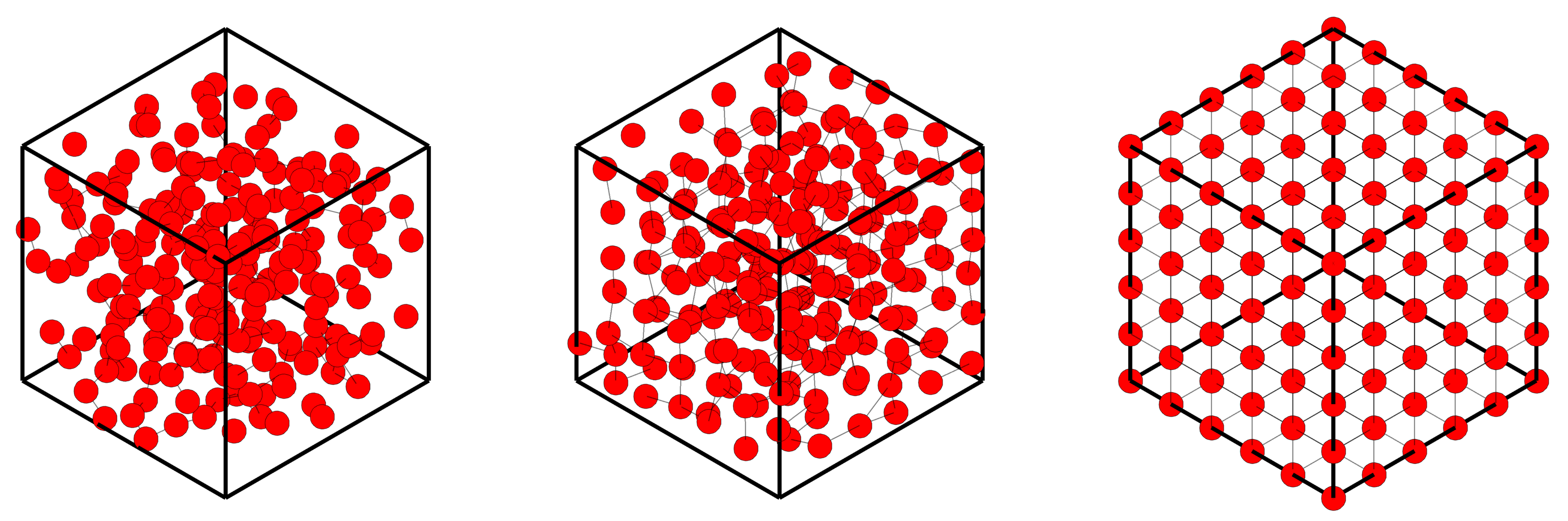


Figure 1. Thermodynamic properties of the system at fixed density $n = 1.0$ for $N = 216$ particles. Both the excess energy and excess heat capacity are shown as functions of the coupling parameter Γ.

- **First-Order Transition**: Sharp jump in $E/N = \langle V \rangle /N$ and peak in C_v at $\Gamma_c = 14.29$ indicate the gas-to-Wigner crystal transition.
- **Wigner Crystal Regime**: For $\Gamma \gg \Gamma_c$, $C_v \approx 3Nk_B$, consistent with Dulong-Petit law for a classical 3D solid.
- **Gas Regime**: For $\Gamma \to 0$, $C_v \to 0$ and Coulomb correlations vanish, yielding a weakly coupled, disordered state.

Particle Configurations

Snapshots show the gas-to-Wigner crystal transition; the observed simple cubic (SC) lattice is a consequence of the cubic geometry and PBC.



(a) $\Gamma = 1.61$. Disordered (gas-like) phase.

(b) $\Gamma = \Gamma_c \approx 14.29$. Onset of local ordering.

(c) $\Gamma = 16119.92$. Wigner crystal showing a SC lattice arrangement.

Figure 2. Snapshots of the $N = 216$ system at density $n = 1.0$, illustrating the system evolution as Γ increases.

Pair-Distribution Function and Structure Factor

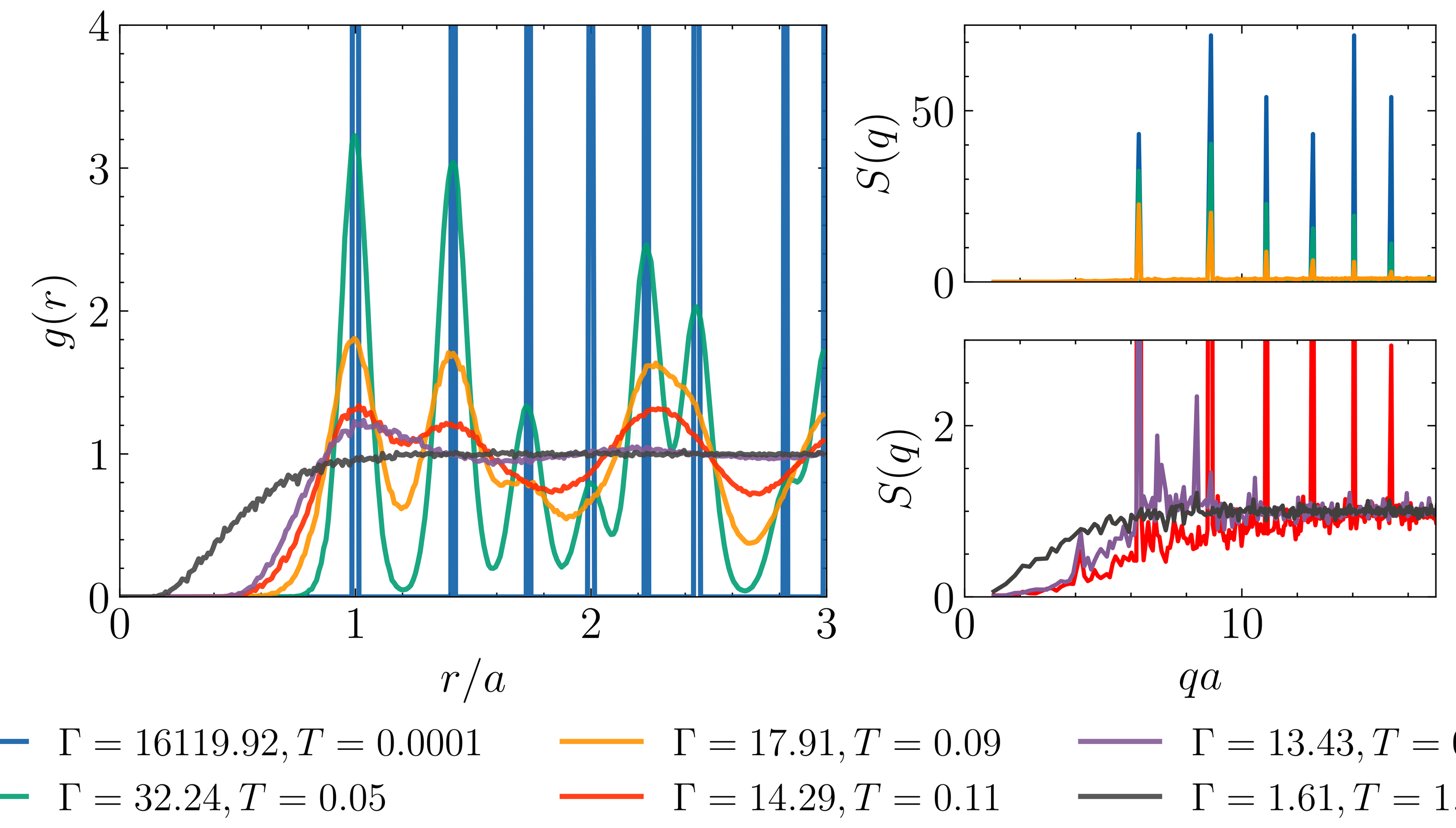


Figure 3. Pair distribution function $g(r)$ and structure factor $S(q)$ for different coupling parameters at density $n = 1.0$.

- **Gas/Liquid Phase** ($\Gamma\downarrow, T\uparrow$): $g(r)$ and $S(q)$ are smooth and featureless $\to$ weak correlations and disordered structure.
- **Crystalline Order** ($\Gamma\uparrow, T\downarrow$): $g(r)$ shows sharp, regularly spaced peaks and $S(q)$ exhibits intense Bragg maxima $\to$ strong correlations and long-range order.
- **Lattice identification**: Peak positions, $r_n/a = \sqrt{n}$ in $g(r)$ and at $q_n a = 2\pi\sqrt{n}$ in $S(q)$, match a SC lattice.

		1st	2nd	3rd	4th	5th	6th	MRE (%)
$g(r)$	Theory	1.000	1.414	1.732	2.000	2.236	2.449	0.2%
	Sim.	**1.005**	**1.415**	**1.735**	**1.995**	**2.235**	**2.445**	
$S(q)$	Theory	6.283	8.885	10.883	12.566	14.050	15.390	0.003%
	Sim.	**6.283**	**8.886**	**10.883**	**12.566**	**14.050**	**15.391**	

Table 1. Comparison of theoretical and simulated peak positions of $g(r)$ and $S(q)$ at $\Gamma = 16119.92$ ($T = 0.0001$) and $n = 1.0$. Last column shows mean relative error (MRE).

- The SC lattice at $\Gamma_c \approx 14.3$ is an artifact of the finite system size ($N = 216$) and the Minimum Image Convention. In the thermodynamic limit, Wigner crystals form a **BCC lattice** [3].

Phase Transition Diagram

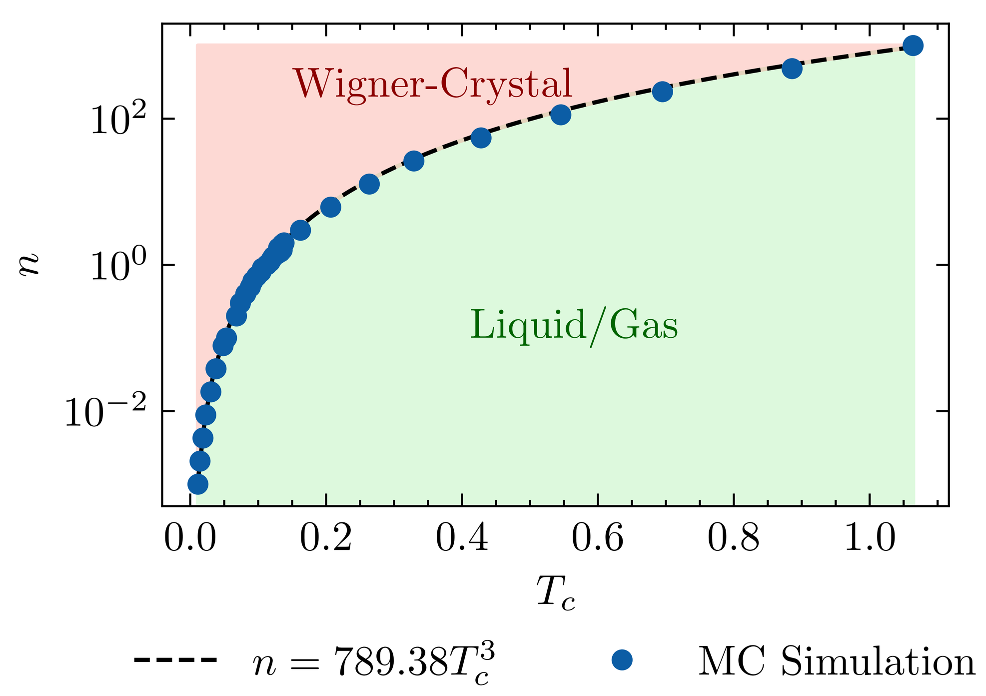


Figure 4. Relation between particle density n and critical temperature T_c. Blue points show Monte Carlo simulation results, and the dashed line theoretical scaling.

- **Gas $\to$ Wigner crystal**: clear boundary between disordered and crystalline phases.
- **Theory vs Simulation:** MC results agree well with the theoretical scaling $n \propto T_c^3$.
- **Critical Coupling:** $\Gamma_{c,\text{fit}} = 14.90$ marks where Coulomb repulsion starts to dominate thermal motion.

Conclusions & Outlook

We successfully observed the phase transition from a Wigner crystal to a gas in a system of $N = 216$ particles at distinct densities n. Structural analysis of $g(r)$ and $S(q)$ at low temperatures confirms a SC lattice, in good agreement with theory. The first-order nature of the transition is evident from a sharp jump in energy and a peak in heat capacity C_v at Γ_c. For $\Gamma \gg \Gamma_c$, $C_v \approx 3Nk_B$ is consistent with crystalline order. These results validate our simulation methodology and qualitative characterization of the phase transition.

Future work includes implementing Ewald summation and a jellium background to properly account for long-range interactions and verify whether a BCC lattice forms. Increasing N will allow investigation of finite-size effects and convergence of the critical coupling Γ_c to quantitatively characterize the system.

References

[1] J. P. Hansen, Phys. Rev. A. **8**, 3096 (1973).

[2] G. S. Demyanov and P. R. Levashov, Phys. Rev. E. **106**, 015204 (2022).

[3] S. G. Brush, H. L. Sahlin, and E. Teller, J. Chem. Phys. **45**, 2102 (1966).

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